```
set nf [open out.nam w]
$ns namtrace-all $nf //Open the NAM trace file

proc finish {}
{
        global ns tracefile nf
        $ns flush-trace
        close $nf
        close $tracefile
        exec nam out.nam &
        exit 0
}
//'finish' procedure

set n0 [$ns node]
set n1 [$ns node]
$ns simplex-link $n0 $n1 1Mb 10ms DropTail // Create your
topology
    - set n0 nodes….
    - make the link between node0 and node1

set tcp0 [new Agent/TCP]
$ns attach-agent $n0 $tcp0
set cbr0 [new Application/Traffic/CBR]
$cbr0 attach-agent $tcp0
set tcpsink0 [new Agent/TCPSink]
$ns attach-agent $n1 $tcpsink0
$ns connect $tcp0 $tcpsink0 // Create your agents
-the nodes will follow the TCP at CBR(Constant Bit Rate)

$ns at 1.0 "$cbr start"
$ns at 3.0 "finish" //Scheduling Events
- $ns at 1.0 start
  and at 3.0 finish

$ns run //starts the simulation
```

This is a basic example of a .TCL file where you create two nodes and observe the transfer of packets according to the Transmission Control Protocol (TCP) at Constant Bit Rate (CBR), which is a traffic generator. In this example, we have used the TCP but other protocols and traffic sources can be used depending on your network architecture.

## Execution of the .TCL file in NS2

.TCL files contain the code of network simulation and can be executed by following the steps given below:

- Once you have installed the NS2 successfully, open a terminal and run the following:

```
/ns filename.TCL.
```

- Once you run that command, *.tr* and *.nam* files will be created in the same directory that contains the .TCL file.

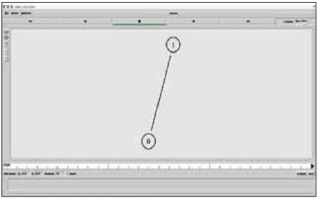

Figure 2: Execution of a simple TCL file

- To interpret the results in the form of animation, you need to just run the command given below:

```
/nam filename.nam
```

- And to see the results in numerical form, you need to open the *.tr* file in the editor.

```
/gedit filename.tr
```

The main aim of this article was to give readers an insight into the architecture and working of NS2 and how it could be used to simulate complex networking problems with simple programming syntax.

As the next step, start with designing a different network architecture consisting of three nodes, write a simple script using the reference given above and check its execution. This would give you a good idea about writing and executing scripts in NS2. Check the 'References' section for more details about simulating your network using Network Simulator 2. END

### References


[1] Introduction to Network Simulator 2, Springer Publications.
[2] The NS2 project page, *http://nsnam.isi.edu/nsnam/index.php/User_Information*
[3] For more examples, visit, *www.nsnam.com*



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